MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

SECOND SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG2DSCECT100 - Essential Concepts in Digital Electronics

Duration: 1½ hours Maximum Marks: 35

Practical Examination – Model Question Paper

Instructions:

- **Demonstrate** thegiven task. (Any one of the tasks listed below)
- **Viva session** (Minimum 5 questions from the Practical Module)
- **Lab Report**

Evaluation Criteria:

- **Viva**: 7 marks
- **Lab Report**: 8 marks
- **Demonstration**: 20 marks

Tasks: *(Choose any one)*

A. Lab experiment using Trainer kit

1. Design and set up a Half Adderusing XOR gates and AND gate
2. Design and set up a Full adder
3. Design and set up a 4:1 Multiplexer
4. Design and set up a 1:4Demultiplexer
5. Design and verify the performance of Serial in Serial out Shift registers
6. Design and verify the performance of Serial in Parallel out Shift Registers
7. Design and set up a 2-bit synchronous counter

B. Lab experiment using Simulation tool

1.Build and verify AND gate using Simulation tool

2.Build and verify OR gate using Simulation tool

3. Build and verify NOT gate using Simulation tool